Gaurav Pandey

Curriculum Vitae

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Education

2022 – Present **Doctor of Philosophy - PhD**, *National Taipei University of Technology*, Taipei, Taiwan

2017 – 2019 **Master's, Mechanical Engineering**, *Chang Gung University*, Taiwan

2012 – 2016 **Bachelor's, Mechanical Engineering**, *A.P.J. Abdul Kalam Technical University*, U.P, India

Publications

1. **Pandey, Gaurav**, et al. "X-ray metrology for advanced semiconductor manufacturing of next-generation logic devices." *Journal of Micro/Nanopatterning, Materials, and Metrology* 25.1 (2026): 010901-010901. **(Journal article)**
2. **Pandey, Gaurav**, et al. "Emerging x-ray metrology techniques with evolving front-end-of-line high-technology IC nodes." *Metrology, Inspection, and Process Control XXXIX* 13426 (2025): 311-318. **(International Conference)**
3. Chein, W. H., **Pandey, G.**, Das, S., & Chen, L. C. (2024, June). Advancements in metrology for advanced semiconductor packaging. In *Optics and photonics for advanced dimensional metrology III* (Vol. 12997, pp. 127-142). SPIE **(International Conference)**

Experience

Mar 2023 – Present **Research Intern**, *Center of Measurement & Standards, ITRI*, Hsinchu, Taiwan
Conduct research on X-ray metrology for in-line critical dimension (CD) measurement in front-end-of-line (FEOL) semiconductor manufacturing.

Sep 2022 – Present **Research Assistant**, *Precision Metrology Lab, NTU*, Taipei, Taiwan
Conduct research focused on understanding optical critical dimension (OCD) metrology and automated optical inspection (AOI) applications.

Feb 2022 – Jun 2022 **Product Design and Development**, *Botlab Dynamics, TBIU, IIT Delhi*, New Delhi
Directed end-to-end design and development of light-show drone fleets across multiple weight classes, scaling from rapid 3D-printed prototypes to mass production.

Aug 2021 – Feb 2022 **Design & Application Engineer**, *Aerogram, TBIU, IIT Delhi*, New Delhi
Responsible for the product development cycle (PDC) of various IoT monitoring devices with targets to meet the criteria of certifications like ATEX and ICAT.

Nov 2019 – Mar 2021 **Asst. Manager-Design & Development**, *Thousand Perfection Melamine Co. Ltd*, Guangdong Province, Dongguan, China
Worked in the R&D division to design and develop compressive mold tools for new products.Part of the team to facilitate methods to monitor and control the critical points of different forming load machinery.

Jul 2016 – Aug 2017 **Design Engineer**, *Sagar Tools and Machines*, Uttar Pradesh, India
Designed and assembled injection-moulding, high-pressure die-casting (HPDC), sheet-metal, and cold/hot-forming tools; developed mould bases, fixtures, and tooling for special-purpose machines. Project work included ASY product development and IKEA product HPDC tool design.

Recognition & Certifications

2026 **Expert Forum Speaker:** "Semiconductor Inspection Technology," 2nd Formosa Conference on Non-Destructive Testing (FCNDT 2026), National Central University, Taoyuan, Taiwan, 15 October 2026.

2024 **Radiation Safety Training for Operators:** Completed an 18-hour course, Chinese Radiation Protection Association, Taiwan, 16–18 April 2024. Certificate No. 1130434.

2018 **CTCI Foundation, Taiwan:** Selected among the top 30 international students for an enterprise-visit program for international graduate students.

Valid to Mar 2027 **NSRRC Facility User Qualification:** Passed the required user assessment and obtained a facility user card, National Synchrotron Radiation Research Center, Taiwan; valid through 20 March 2027.

Technical **ANSYS Fluent / CFD:** Training and certification, CADMEN Taiwan Auto-Design Co.

SolidWorks and AutoCAD: Training and certification, Prestantia Creation Pvt. Ltd.

Advanced Mould Tool Design: Training in plastic injection, high-pressure die casting (HPDC), and blow moulds, Tech Labs.

3D Printing and Unmanned Aerial Vehicles: Summer Immersion Program, Providence University, Taiwan.

Fundamentals of IoT and Data Science: Course certification.

Internships & Training

1 month **Industrial Intern**, *Tata Motors (Shri Ram Foundry)*, Uttarakhand, India
Supported maintenance checks of pneumatic and hydraulic brakes and assembly-line jacks; studied combustion-engine layouts and automotive casting processes.

1 month **Power Plant Simulation Intern**, *STEAG Energy Services India Pvt. Ltd.*, Uttar Pradesh, India
Estimated power-plant efficiency at 150, 200, and 250 MW loads using basic simulation models in Ebsilon Professional.

TFBE **Research Intern – Microbial Fuel Cells**, *National Ilan University*, Taiwan
Reviewed applications of artificial intelligence (AI) and machine learning (ML) to microbial fuel cell (MFC) optimization and explored novel AI approaches for MFC development.

Project and Academic Training

Teaching Teaching Assistant for undergraduate Fluid Dynamics, Chang Gung University (CGU), Taiwan, 2018–2019 (one year).

Design Freelance CAD and product-design work for STM Pvt. Ltd., PUW, and online clients, including thermosetting and thermoplastic mould tools.

Prototyping Developed Arduino-based automation for hydroponic and aquaponic systems, supporting exploration of modern agricultural techniques.

Community Participated in RedX CGU, a community associated with the MIT Media Lab, contributing to projects addressing social needs.

Technical Skills

Programming Python, MATLAB, C, Arduino

CAD / CAE SolidWorks, Siemens NX, AutoCAD, Proteus, ANSYS, QForm, Ebsilon Professional

Languages

Hindi, English, Mandarin